



# DeepSea Guardian

AI-Powered Ocean Monitoring & Marine Life Protection Platform

Complete Project Guide & System Walkthrough

AI Chemical Spill Detection

Autonomous Drone Dispatch

6-Hour Pollution Forecast

Marine Species Vision AI

Live Ocean Map

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**Repository:** <https://github.com/tiwariraman884/DeepSee.git>

# DeepSea Guardian (DeepSee) — Complete Project Guide

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AI-Powered Ocean Monitoring & Marine Life Protection Platform

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## 1. Introduction: What is DeepSea Guardian?

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**DeepSea Guardian (DeepSee)** is an intelligent ocean protection platform. It works like an automated mission control center for our oceans.

By combining smart ocean sensors, Artificial Intelligence (AI), and autonomous underwater drones, DeepSea Guardian continuously watches the ocean to:

- **Detect Water Pollution & Chemical Spills** instantly when they happen.
- **Dispatch Autonomous Underwater Drones** to the exact spill location for live video inspection.
- **Forecast Pollution Movement** up to 6 hours into the future.
- **Identify Marine Animals** using an AI-powered smart camera scanner.
- **Show Everything on a Live Map** so ocean scientists and conservation teams can take immediate action.

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## 2. Why is this Project Important?

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Our oceans cover over 70% of the planet, but more than 80% of deep-sea areas are completely unmonitored.

### The Challenges:

1. **Late Pollution Alerts:** Oil spills, chemical leaks, and illegal dumping often go unnoticed for days or weeks.
2. **Harm to Marine Life:** Endangered animals like sea turtles, whales, and coral reefs get damaged before help arrives.
3. **Expensive Manual Inspections:** Sending human research ships to check the deep sea is slow and costs thousands of dollars per hour.

### How DeepSea Guardian Solves This:

DeepSea Guardian provides **24/7 automated monitoring**. The moment dirty water or a chemical leak is detected by an ocean sensor, the AI flags the emergency and dispatches an unmanned robot drone within seconds.

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### 3. How the System Works (End-to-End Flow)

The entire project works in **4 simple steps**:

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[1. Ocean Sensors & Buoys]
    | (Measure pH, Oxygen, Temperature, Turbidity)
    ▼
[2. AI Brain & Cloud Backend]
    | (Analyzes readings & detects abnormal spikes)
    ▼
[3. Live Mission Control Dashboard]
    | (Displays live alerts, maps, and forecasts)
    ▼
[4. Autonomous Drones Dispatched]
    | (Investigate hotspots and scan marine life)
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1. **Step 1 — Collecting Ocean Data:** Sensor buoys floating in the ocean measure water quality (pH level, dissolved oxygen, water clarity/turbidity, temperature, and salt levels).
2. **Step 2 — AI Brain Analysis:** The data is sent to an AI algorithm (Isolation Forest) that checks whether the water is clean or contaminated.
3. **Step 3 — Alert on the Dashboard:** If a chemical leak or pollution is found, a bright red alarm appears on the web dashboard with full details.
4. **Step 4 — Drone Action:** The operator clicks one button, and an autonomous underwater drone (**AquaDrone-Alpha**) is dispatched to inspect the hotspot.

### 4. Key Features Explained Simply

#### Feature 1: AI Anomaly & Chemical Spill Detector 🧪

- **What it does:** Checks water measurements in real time to spot pollution.
- **How it works:**
- **Clean Water:** Normal ocean water has a balanced pH (around 8.1), high oxygen (5+ mg/L), and clear water (low turbidity).
- **Chemical Spill:** If an oil leak or chemical spill occurs, the water becomes acidic (pH drops to 6.5), oxygen drops, and water becomes murky (turbidity rises).
- **In the App:**
- Users can test two quick scenarios: "**Clean Ocean Baseline**" or "⚠️ **Trigger Chemical Spill Scenario**".

- When the chemical spill is triggered, the AI immediately flags a **Critical Anomaly** and unlocks the **Drone Dispatch** button.
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## Feature 2: Autonomous Underwater Drone Dispatch 🚁

- **What it does:** Sends a robotic underwater drone to inspect dangerous or deep locations where humans cannot quickly reach.
  - **In the App:**
  - When a pollution event is detected, clicking "**Dispatch Nearest Drone for Emergency Inspection**" instantly assigns **AquaDrone-Alpha**.
  - The system shows the drone's status, destination sector, and estimated arrival time (ETA: ~3 minutes).
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## Feature 3: AI Pollution Spread Forecast 📊

- **What it does:** Predicts how far and how strong a pollution spill will spread over the next 6 hours.
  - **Why it matters:** Ocean currents and waves carry toxic waste. Knowing the future path allows teams to protect nearby beaches and coral reefs before the spill reaches them.
  - **In the App:**
  - The user sets the current pollution severity (from 1 to 10) and the trend (Increasing, Stable, or Decreasing).
  - Clicking "**Generate 6h Forecast**" draws a 6-hour prediction graph (+1h, +2h, +3h, +4h, +5h, +6h).
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## Feature 4: AI Marine Species Identifier (Smart Vision) 🐠

- **What it does:** Identifies sea animals from photos taken by drone cameras or uploaded by users.
  - **In the App:**
  - Located on the **Biodiversity** page.
  - Users can click **1-Click Test Samples** (such as *Hawksbill Sea Turtle*, *Blue Whale*, *Clownfish*, *Hammerhead Shark*) or drag and drop any fish photo.
  - The AI model scans the photo and instantly displays the **Species Name** along with an **AI Confidence Score (%)**.
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Feature 5: Interactive Ocean Map & Ecosystem Time Machine 🗺️ ⏳

- **Interactive Map:** Displays a worldwide dark-themed ocean map with colored pins showing active pollution hotspots, sensor buoys, and drone locations.
- **Ecosystem Time Machine:** A timeline slider that lets users travel forward in time (**Today, 1 Month, 6 Months, 1 Year, 5 Years**) to see projected ocean health changes and biodiversity impact.

Feature 6: AI Ocean Assistant Chatbot 💬

- An integrated AI assistant ready to answer questions about ocean health, explain recent alerts, and suggest environmental cleanup protocols in natural language.

5. Technology Behind the Project

The project is built using modern, fast, and reliable tools:

| Component               | Technology Used                      | What It Does                                                                                          |
|-------------------------|--------------------------------------|-------------------------------------------------------------------------------------------------------|
| Frontend (Website)      | Next.js 15 & React 19                | Creates the fast, interactive, and responsive Mission Control dashboard.                              |
| Styling                 | Tailwind CSS                         | Gives the app a modern dark ocean theme with smooth glassmorphism effects.                            |
| Backend Server          | Node.js & Express (TypeScript)       | Handles data traffic, routes API requests, and connects the website to the AI brain.                  |
| Database                | SQLite                               | Stores drone missions, past alerts, and sensor histories locally and reliably.                        |
| Artificial Intelligence | Python (Scikit-Learn, NumPy, Pillow) | Runs the machine learning models for anomaly detection, spread prediction, and animal classification. |
| Interactive Maps        | Leaflet & OpenStreetMap              | Powers the live geospatial ocean tracking map.                                                        |

## 6. Easy Step-by-Step User Guide (How to Demo)

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### Step 1: Open the Mission Control Dashboard

1. Open your browser and go to `http://localhost:3000`.
2. You will see the main dashboard with live ocean health scores, active drone counters, and the ocean map.

### Step 2: Test the Chemical Spill & Drone Dispatch

1. Scroll down to the **AI Anomaly Detector** card.
2. Click the glowing red button: `⚠️ Trigger Chemical Spill Scenario`.
3. Watch the AI analyze the data and display the "**CRITICAL ANOMALY DETECTED**" alarm.
4. Click `🚁 Dispatch Nearest Drone for Emergency Inspection`.
5. See the real-time confirmation showing `AquaDrone Alpha Dispatched • ETA: 3m 45s`.

### Step 3: Test the 6-Hour Spread Forecast

1. Look at the **AI Spread Forecast** card right next to the anomaly detector.
2. Set the severity slider to **7.5** and choose **Trend: Increasing**.
3. Click `Generate 6h Forecast`.
4. A chart will appear displaying predicted pollution severity for the next 6 hours.

### Step 4: Test the Marine Species AI Scanner

1. Click "**Biodiversity**" in the left sidebar (or navigate to `http://localhost:3000/species`).
2. At the top, you will see the **AI Species Identifier**.
3. Click any sample button (like `🐢 Hawksbill Turtle` or `🐠 Clownfish`), or upload your own fish photo.
4. The AI will scan the picture and display the identified species with confidence percentage.

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## 7. Project Summary & Key Benefits

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- ⚡ **Instant Response:** Reduces pollution detection time from days to under 1 second.
- 🚁 **Autonomous Robotics:** Dispatches drones automatically to reduce human risk and survey costs.
- 📈 **Proactive Prevention:** Forecasts pollution trajectories before they reach coastlines.
- 🐠 **Biodiversity Protection:** Tracks and preserves endangered sea creatures with smart vision.

-  **Modern & Fast:** Built with Next.js 15, responsive across desktop, tablet, and mobile devices.
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*DeepSea Guardian — Protecting our Oceans with Artificial Intelligence.* 